DO NOW — RECALL

Lesson 048 · C# List<T>: Add, Insert and Remove

Name: _____ Class: _____ Date: _____

1. What does Array.Length give?

2. What is zero-based indexing?

3. What is aliasing between references?

4. A rising-edge D-type flip-flop starts with Q=0. D rises to 1 and then returns to 0 entirely between two rising clock edges. Explain why Q remains 0.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Number of elements in the array.

2. Model answer

First item is at index zero.

3. Model answer

Two references identify the same object.

4. Model answer

No rising clock edge samples D while it is 1. The flip-flop holds Q=0 between active edges and samples 0 at the next rising edge.

By the end, you can…

• List<T> is a resizable sequence. Add appends; Insert inserts at a specified index and shifts following items.

• RemoveAt removes an item by index; Remove(value) removes the first matching value instead.

• Unlike an array, List<T> changes Count as items are inserted/removed. An index must still be within the valid bounds.

Vocabulary — know these terms

List<T> — Resizable generic sequence.

Insert — Adds at a specified index.

RemoveAt — Removes item by index.

Count — Number of items currently stored.

NEW LEARNING

Read, annotate and explain

Observe

List<T> is a resizable sequence. Add appends; Insert inserts at a specified index and shifts following items.

Explain

RemoveAt removes an item by index; Remove(value) removes the first matching value instead.

Apply

Unlike an array, List<T> changes Count as items are inserted/removed. An index must still be within the valid bounds.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Trace a list after inserting at the start, then removing the last element.

Trace and justify

After Insert: [1,2,5,7]; RemoveAt(3) removes 7; final [1,2,5].

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Trace Insert(1,6) then RemoveAt(2) for [4,8,12]. [6]

Question 2

Explain the difference between Remove(5) and RemoveAt(1) for [2,5,5,9]. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

For [1,2,3], write C# that inserts 0 at start, appends 4 and outputs Count; state why List<T> suits a growing collection. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

RemoveAt(5) means remove the value 5.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.